

Optics

Quiz 2 | Notes and worked solutions

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N02 Integration techniques

Math toolbox for the following notes and solutions

Euler identities and a basic integral

$$\cos ax = \frac{e^{iax} + e^{-iax}}{2}, \quad \sin ax = \frac{e^{iax} - e^{-iax}}{2i}$$

$$\int_{-L}^L e^{-iqx} dx = \frac{e^{-iqL} - e^{iqL}}{-iq} = 2L \operatorname{sinc}(qL)$$

Use the limit $2L$ when $q = 0$. This one integral produces most windowed sine, cosine and rectangular-pulse transforms.

$$\sin^2 ax = \frac{1 - \cos 2ax}{2}, \quad 2\cos A \cos B = \cos(A + B) + \cos(A - B)$$

Even and odd functions

$$\int_{-L}^L g(x) dx = 2 \int_0^L g(x) dx \quad \text{if } g \text{ is even}$$

$$\int_{-L}^L g(x) dx = 0 \quad \text{if } g \text{ is odd}$$

For real even f , the sine part of its transform vanishes. For real odd f , the cosine part vanishes. Check the symmetry about the chosen origin, not about the visual centre of a shifted graph.

Integration by parts

$$\int u dv = uv - \int v du$$

$$\int x \cos ax dx = \frac{x \sin ax}{a} + \frac{\cos ax}{a^2}$$

Choose the polynomial as u : differentiating it reduces the power. In a definite integral, evaluate the boundary term too. For x^2 , integrate by parts twice.

Substitution and complex amplitudes

$$y = ax \Rightarrow dx = dy/a, \quad y = x - x_0 \Rightarrow x = y + x_0$$

$$|Ae^{i\phi}|^2 = |A|^2, \quad |A + B|^2 = |A|^2 + |B|^2 + 2\operatorname{Re}(AB^*)$$

For negative scale a , reversing the integration limits gives $1/|a|$. Add complex amplitudes before squaring: the cross term is interference.

F04 Fourier transforms and scaling of delta functions

Compiled Q4; Delta Function problem

Find the Fourier transform of $\delta(ax)$. Prove $\delta(ax, by) = \delta(x, y)/|ab|$. Find the Fourier transform of $\delta(x/a - 1)$. Assume a and b are nonzero real constants.

Use the sifting and scaling rule $\delta(ax) = \delta(x)/|a|$, with the forward-transform convention given on page 1.

$$\mathcal{F}\{\delta(ax)\} = \frac{1}{|a|} \int_{-\infty}^{\infty} \delta(x) e^{-ikx} dx = \frac{1}{|a|}$$

$$\delta(ax, by) = \delta(ax)\delta(by) = \frac{1}{|a||b|} \delta(x)\delta(y) = \frac{1}{|ab|} \delta(x, y)$$

Rewrite the shifted delta before transforming it; its impulse is at $x = a$ and its area is $|a|$.

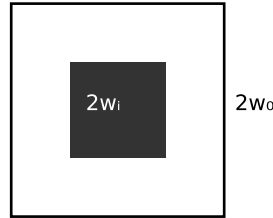
$$\delta\left(\frac{x}{a} - 1\right) = |a| \delta(x - a)$$

$$\mathcal{F}\left\{\delta\left(\frac{x}{a} - 1\right)\right\} = |a| e^{-ika}$$

D01 Square aperture with a square obstruction

Tutorial 3 p2; Lipson 8.2

Consider the square aperture below. The black square is opaque and the white region is transparent. (a) Write the transmission function. (b) Find the amplitude and intensity distributions of its Fraunhofer diffraction pattern, ignoring common prefactors. Assume unit-amplitude, normally incident plane-wave illumination.



Let the outer clear square have side $2w_0$ and the centered opaque square have side $2w_i$. The transmitted field is the outer square minus the blocked square. A centered square of side $2w$ transforms to the product of two sinc functions.

$$t(x, y) = \text{rect}(x/(2w_0))\text{rect}(y/(2w_0)) - \text{rect}(x/(2w_i))\text{rect}(y/(2w_i))$$

$\text{rect}(s)=1$ for $|s|<1/2$ and 0 outside; $\text{sinc}(s)=\sin(s)/s$.

$$T_w(u, v) = \int_{-w}^w e^{-iux} dx \int_{-w}^w e^{-ivy} dy$$

$$T_w(u, v) = (2w \text{sinc}(wu))(2w \text{sinc}(wv))$$

$$A(u, v) \propto 4w_0^2 \text{sinc}(w_0u) \text{sinc}(w_0v) - 4w_i^2 \text{sinc}(w_iu) \text{sinc}(w_iv)$$

$$I(u, v) \propto |A(u, v)|^2$$

At $u=v=0$ the amplitude is proportional to the transmitted area, $4(w_0^2 - w_i^2)$. The obstruction subtracts its own square-aperture spectrum coherently.

C01 Fourier series and transform pairs

Cheat sheet 1 of 3 • sinc $u = \sin u / u$; all widths below are positive.

$$f = \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos nKx + b_n \sin nKx), \quad K = 2\pi/P$$

$$a_n = \frac{2}{P} \int_P f \cos nKx \, dx, \quad b_n = \frac{2}{P} \int_P f \sin nKx \, dx, \quad a_0 = \frac{2}{P} \int_P f \, dx$$

$$f = \sum_{n=-\infty}^{\infty} c_n e^{inKx}, \quad c_n = \frac{1}{P} \int_P f(x) e^{-inKx} \, dx$$

Even: cosine coefficients only. Odd: sine coefficients only. DC = $a_0/2 = c_0$. At a jump take the midpoint value.

$$F(k) = \int f(x) e^{-ikx} \, dx, \quad f(x) = \frac{1}{2\pi} \int F(k) e^{ikx} \, dk$$

$$\text{rect}(x/a) \longleftrightarrow a \text{sinc}(ka/2)$$

$$\text{tri}(x/a) \longleftrightarrow a \text{sinc}^2(ka/2)$$

$$e^{-x^2/(2\sigma^2)} \longleftrightarrow \sqrt{2\pi}\sigma e^{-\sigma^2 k^2/2}$$

$$\delta(x - x_0) \longleftrightarrow e^{-ikx_0}, \quad 1 \longleftrightarrow 2\pi\delta(k)$$

$$\cos k_1 x \longleftrightarrow \pi[\delta(k - k_1) + \delta(k + k_1)]$$

$$\sin k_1 x \longleftrightarrow \frac{\pi}{i}[\delta(k - k_1) - \delta(k + k_1)]$$

$$\sum_n \delta(x - nd) \longleftrightarrow \frac{2\pi}{d} \sum_m \delta(k - 2\pi m/d)$$

$$\delta(ax) = \delta(x)/|a|, \quad \int f(x) \delta(x - x_0) \, dx = f(x_0)$$

Fast checks: $F(0)$ =area; real f gives $F(-k)=F^*(k)$; increasing width narrows the spectrum. Do not replace angular frequency k with cycles/length without $k=2\pi\nu$.